A commentary by Friedrich Boettner, MD, is linked to the online version of this article.

Outcomes Following Direct Anterior Approach Total Hip Arthroplasty

A Contemporary Multicenter Study

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Background: The direct anterior approach (DAA) is a popular approach for primary total hip arthroplasty (THA). However, the contemporary outcomes for DAA THA need further elucidation. Therefore, we aimed to describe implant survivorship, complications, and clinical outcomes after DAA THA.

Methods: From our multi-institutional total joint registry, 3,184 patients who had undergone 3,698 primary DAA THA between 2010 and 2019 were identified. The identified patients had a mean age of 65 years and a mean body mass index (BMI) of 29 kg/m^2, and 53% of patients were female. The indications for revision and reoperation and the incidence of complications were collected and analyzed. Potential risk factors, including age, sex, BMI, and high-volume compared with low-volume operating surgeons, were examined. Descriptive statistics and Kaplan-Meier survivorship with Cox regression analyses were performed.

Results: At 10 years following primary DAA THA, the cohort had 96% (95% confidence interval [CI], 95% to 98%) survivorship free of any revision and 94% (95% CI, 92% to 96%) survivorship free of any reoperation. The leading indications for revision were periprosthetic joint infection (PJI) (n = 24; 5-year cumulative incidence, 0.93% [95% CI, 0.6% to 1.5%]), periprosthetic fracture (n = 20; 5-year cumulative incidence, 0.62% [95% CI, 0.4% to 1.0%]), and aseptic loosening (n = 14; 11 femoral, 3 acetabular; 5-year cumulative incidence, 0.84% [95% CI, 0.5% to 1.5%]). A BMI of ≥40 kg/m^2 was found to be significantly associated with PJI (hazard ratio [HR], 6.4; $p < 0.001$), reoperation (HR, 3.5; $p < 0.001$), and nonoperative complications (HR, 2.3; $p = 0.018$). Survivorship free of recurrent instability was 99.6% (95% CI, 99.4% to 99.8%) at 5 and 10 years, and the cumulative incidence of revision for instability was 0.14% at 5 years.

Conclusions: In one of the largest published series to date, survivorship following DAA THA was satisfactory at early to intermediate follow-up. The leading indications for revision were PJI, periprosthetic fracture, and aseptic loosening. Instability after DAA THA was uncommon and infrequently led to revision. As a note of caution, a BMI of ≥40 kg/m^2 was identified as a risk factor for adverse outcome after DAA THA.

Level of Evidence: Therapeutic Level IV. See Instructions for Authors for a complete description of levels of evidence.

Direct anterior approach (DAA) total hip arthroplasty (THA) is now well established as a successful approach for primary THA[1,2]. The utilization of this approach has increased, and a recent poll of the American Association of Hip and Knee Surgeons (AAHKS) found that nearly 50% of members are using the DAA, a sharp increase over the last decade[1]. Although early experience with this approach was marred by high complication rates and an associated learning curve[3-8], many surgeons now have had structured training in DAA THA. Contemporary DAA THA is also facilitated by approach-specific instrumentation and implants.

Substantial effort has been employed in comparing outcomes between surgical approaches in primary THA[2,8-14]. Although many approaches have been successfully utilized, there has been some indication that the reasons for failure after DAA THA may be distinct from those seen with other approaches[14-19]. However,

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much of the existing data are predicated on small series of patients or have limited follow-up.

Given technique, implant, and instrumentation advances, contemporary outcomes after DAA THA are worth exploring. Therefore, we sought to examine implant survivorship, complications, and clinical outcomes after DAA THA.

Materials and Methods

After institutional review board approval, our multi-institutional total joint registry was used to identify all patients who underwent primary DAA THA from 2010 to 2019. Patients who had surgery at 1 of 3 distinct academic institutions within the same health-care system were included. This study identified and included 3,698 primary DAA THAs performed in 3,184 patients. Patients were included regardless of the indication for the surgical procedure, and 23 surgeons performed cases during the study period. Nine of the 23 surgeons performed >50 cases during the study period, and these 9 surgeons accounted for nearly all (96%) of the cases included in the study. The remaining 14 surgeons collectively performed 158 included cases (4%). All included cases were performed on a specialized table (Hana Orthopedic Surgery Table; Mizuho OSI) with intraoperative use of fluoroscopy.

Patients were followed prospectively by our total joint registry at 1, 2, 5, and 10 years postoperatively. Revisions, reoperations, and complications were captured and were confirmed using manual chart extraction. Clinical outcomes were quantified using the Harris hip score (HHS) as previously described[20]. Periprosthetic joint infection (PJI) was diagnosed using standardized criteria[21]. Superficial wound-healing issues were considered a complication if clinical intervention was required (i.e., greater-than-normal wound care, antibiotics, or irrigation and debridement).

The mean patient age was 65 years, the mean body mass index (BMI) was 29 kg/m^2, and 53% of patients were female. The specific indications for primary DAA THA were osteoarthritis (83%), posttraumatic arthritis (4%), osteonecrosis (4%), inflammatory arthropathy (2%), or other (6%). The majority (96%) of patients had no history of undergoing a surgical procedure on the operatively treated extremity, 95% of patients had an American Society of Anesthesiologists (ASA) class of 2 or 3, and the mean age and severity-weighted Charlson Comorbidity Index (and standard deviation) was 3.7 ± 2.9 (Table I)[22]. The mean operative time was 92 ± 27 minutes. Cemented stems were used in 92 cases (2.5%), and uncemented stems were used in 3,606 cases (97.5%). Collared stems were used in 1,359 cases (37%), and collarless stems were used in 2,339 (63%). The femoral head sizes utilized can be seen in Table I.

In 51 cases (1.4% of the 3,698 included THAs), the patient underwent revision within the first 2 years; in 59 cases (1.6%), the patient died within 2 years; and in 1,616 cases (44%), the patient had <2 years of follow-up. Among the patients who were alive and had not undergone a revision at 2 years (1,972 hips), the median and mean follow-up were 4 years (range, 2 to 11 years).

Statistical Analysis

Descriptive statistics were calculated and reported as the mean and the standard deviation or as frequencies and percentages, as appropriate. The rates of revision, reoperation, and complications and the specific indications for revision were reported as 5-year cumulative incidences. When analyzing specific indications for revision, a revision for an indication different from that being analyzed was accounted for as a competing risk. Survivorship was assessed utilizing the Kaplan-Meier method with end points of any revision, any reoperation, any complication, recurrent instability, periprosthetic fracture requiring reoperation, and PJI. All hips were included in survivorship analyses and were followed individually (even in patients with bilateral DAA THA) until revision, patient death, or final follow-up. Cox regression analysis was performed to analyze select risk factors for their association with examined outcomes. For this analysis, we dichotomized surgeon volume at 75 cases over the study period. Age and BMI were analyzed both as continuous variables and as dichotomized variables. Age was dichotomized at ≥65 years and

TABLE I Patient Demographic Data and Operative Information (N = 3,698 Cases)

Female sex*	1,953 (52.8%)
Age at surgery† *(yr)*	64.9 ± 11.4
Laterality*	
Left side	1,716 (46.4%)
Right side	1,982 (53.6%)
BMI† *(kg/m^2)*	29.1 ± 5.8
ASA class*	
1	178 (4.9%)
2	2,385 (65.3%)
3	1,066 (29.2%)
4	22 (0.6%)
Missing	47
Charlson Comorbidity Index, age and severity-weighted†	3.7 ± 2.9
Total no. of prior hip operations*	
0	3,547 (95.9%)
1	94 (2.5%)
≥2	57 (1.5%)
Stem fixation*	
Uncemented	3,606 (97.5%)
Cemented	92 (2.5%)
Femoral head size*	
22 mm	27 (0.7%)
28 mm	80 (2.2%)
32 mm	774 (20.9%)
36 mm	2,718 (73.5%)
40 mm	79 (2.1%)
44 mm	3 (0.08%)
Dual mobility	17 (0.5%)

*The values are given as the number of patients, with the percentage in parentheses. †The values are given as the mean and the standard deviation.

BMI was dichotomized at 40 kg/m^2. These were multivariable analyses controlling for potential confounders (age, sex, BMI, age and severity-weighted Charlson Comorbidity Index, and surgeon volume) where appropriate. The impact of collared stems compared with non-collared stems was also assessed with Cox regression with an outcome of reoperation for periprosthetic fracture. The proportional hazards assumption was examined for each Cox regression graphically, via plotting of the scaled Schoenfeld residuals against event times, and by calculating a formal score test. The proportional hazards assumption was met for all Cox models. The Benjamini-Hochberg method was used to adjust p values in order to control for the false discovery rate. All analyses were conducted using SAS (version 9.4M6; SAS Institute) and R (version 4.2.2; The R Foundation).

Results

Survivorship Free of Revision and Reoperation

Among the 3,698 DAA THAs, there were 123 hips that underwent reoperation, of which 65 required component revision (Table II). Survivorship free of any revision was 98% (95% confidence interval [CI], 97.9% to 98.8%) at 2 years, 97% (95% CI, 96.5% to 98.1%) at 5 years, and 96% (95% CI, 94.6% to 97.9%) at 10 years postoperatively (Fig. 1). The indications for revision were PJI (24 [37% of all revisions]), periprosthetic fracture (20 [31%]), aseptic loosening (14 [22%]; 11 femoral and 3 acetabular), recurrent instability (4 [6%]), trunnion corrosion (n = 2 [3%]), and leg-length discrepancy (1 [1.5%]) (Table II). At 5 years, the cumulative incidence of revision for these indications was 0.93% (95% CI, 0.59% to 1.48%) for PJI, 0.62% (95% CI, 0.39% to 1.01%) for periprosthetic fracture, 0.84% (95% CI, 0.46% to 1.53%) for aseptic loosening (femoral loosening, 0.67%; acetabular loosening, 0.17%), 0.14% (95% CI, 0.05% to 0.37%) for instability, 0.13% (95% CI, 0.03% to 0.57%) for trunnion corrosion, and 0.04% (95% CI, 0.01% to 0.31%) for leg-length discrepancy. Patients with a BMI of ≥40 kg/m^2 had lower revision-free survivorship relative to patients with a BMI of <40 kg/m^2 (98% compared with 92% at 5 years, and 96% compared with 92% at 10 years), but this was not significant after adjusting for multiple comparisons (hazard ratio [HR], 2.73 [95% CI, 1.2 to 6.4]; p = 0.06). There were 6 revisions in the 137 patients with a BMI of ≥40 kg/m^2: 5 for PJI and 1 for a Vancouver B2 periprosthetic femoral fracture. Survivorship free of PJI was lower in patients with a BMI of ≥40 kg/m^2 compared with patients with a BMI of <40 kg/m^2: 92.5% compared with 99.4% at 5 years and 92.5% compared with 98.9% at 10 years (HR, 6.41 [95% CI, 2.43 to 16.91]; p < 0.001). Patient sex (p = 0.47), patient age (p = 0.47), and surgeon volume (p = 0.54) were not associated with revision-free survivorship (Table III).

Survivorship free of any reoperation was 97% (95% CI, 96.2% to 97.4%) at 2 years, 95% (95% CI, 94.5% to 96.4%) at 5 years, and 94% (95% CI, 91.5% to 95.6%) at 10 years postoperatively (Fig. 2). The 61 non-revision reoperations included superficial irrigation and debridement for wound complications (42 [69%]), arthroscopic iliopsoas tenotomy for symptomatic impingement (9 [15%]), open reduction and internal fixation (ORIF) of periprosthetic femoral fracture (6 [10%]), removal of symptomatic heterotopic ossification (2 [3%]), and other (2 [3%]) (Table II). Female patients (HR, 1.8 [95% CI, 1.2 to 2.6]; p = 0.01) were significantly more likely to require reoperation relative to male patients, and patients with a BMI of ≥40 kg/m^2 (HR, 3.5 [95% CI, 2.0 to 6.2]; p < 0.001) were significantly more likely to require reoperation relative to patients with a BMI of <40 kg/m^2. An increase in BMI by 5 kg/m^2 was also significantly associated with increased reoperation (HR, 1.34 [95% CI, 1.2 to 1.5, p < 0.001). Patient age (analyzed continuously or dichotomized) and surgeon volume were not significantly associated with reoperation (Table III).

Survivorship free of reoperation or revision for periprosthetic femoral fracture was 98.8% (95% CI, 97.9% to 99.7%) at 10 years. Among periprosthetic fractures prompting reoperation within 2 years, 21 occurred in stem designs without a collar and 3 occurred in stems with a collar. Consequently, stems without a collar had a significantly higher hazard of periprosthetic fracture requiring reoperation within 2 years postoperatively (HR, 4.3 [95% CI, 1.3 to 14.6]; p = 0.017). Of note, the proportional hazards assumption was examined and met for all Cox models.

TABLE II Revisions, Reoperations, and Complications After DAA THA*

Revisions	65 (2.7%)
For PJI	24 (37%)
For periprosthetic fracture	20 (31%)
For aseptic loosening	14 (22%)
Femoral aseptic loosening	11 (17%)
Acetabular aseptic loosening	3 (5%)
For recurrent instability	4 (6%)
For trunnion corrosion	2 (3%)
For leg-length discrepancy	1 (2%)
Any reoperations†	61 (2%)
Superficial irrigation and debridement	42 (69%)
For iliopsoas impingement	9 (15%)
ORIF of periprosthetic femoral fracture	6 (10%)
Heterotopic ossification excision	2 (3%)
Other	2 (3%)
Nonoperative complications†	165 (5.0%)
Periprosthetic fracture	59 (36%)
Intraoperative	30 (51%‡)
Postoperative	29 (49%‡)
Wound complication	56 (34%)
Heterotopic ossification§	16 (10%)
Venous thromboembolism	14 (8%)
Recurrent instability	8 (5%)
Lateral femoral cutaneous neuropathy	6 (4%)
Other	6 (4%)

*The values are given as the number of cases, with the percentages in parentheses. ORIF = open reduction and internal fixation. †The overall revision, reoperation, and complication percentages represent cumulative incidence at 5 years. ‡These are the percentages of the periprosthetic fractures. §Brooker grades 3 and 4.

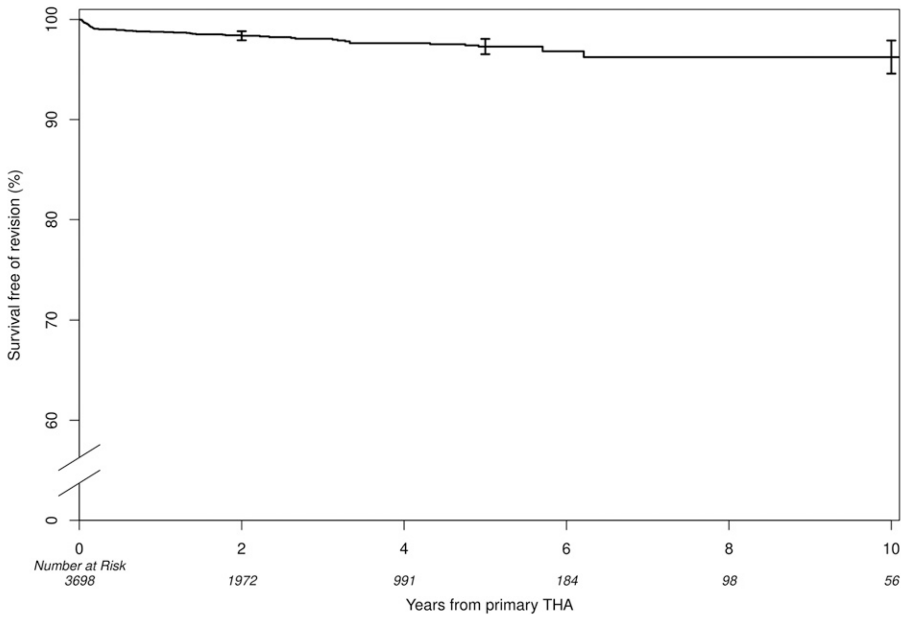

Fig. 1
Estimated survivorship free of any revision was 98% at 2 years, 97% at 5 years, and 96% at 10 years postoperatively. The error bars indicate the 95% CIs.

Complications

Survivorship free of nonoperative complications after DAA THA was 95% (95% CI, 94.7% to 96.1%) at 2 years, 95% (95% CI, 94.3% to 95.9%) at 5 years, and 94% (95% CI, 92.2% to 95.6%) at 10 years (Fig. 3). There were a total of 165 complications not requiring reoperation or revision surgery. These included 59 nonoperatively treated periprosthetic fractures (36%; 30 intraoperative and 29 postoperative fractures); 56 wound complications managed with wound care with or without antibiotics (34%); 16 cases of Brooker grade-3 or 4 heterotopic ossification (10%); 14 cases of venous thromboembolism (8%); 8 recurrent instabilities (5%); 6 cases of transient symptomatic lateral femoral cutaneous neuropathy (4%); and 6 cases of other complications (4%) (Table II). Survivorship free of recurrent instability was 99.6% (95% CI, 99.4% to 99.8%) at 5 and 10 years. Of the 12 recurrent instabilities, 8 were managed nonoperatively and 4 required revision THA. Eleven of the 12 recurrent instabilities occurred within 4 months of the surgical procedure, and all were within the first postoperative year. The direction of recurrent instability was known for 10 patients, with 8 hips dislocating anteriorly and 2 hips dislocating posteriorly. The femoral head size among the patients who had recurrent instabilities was either 32 mm (6) or 36 mm (6).

The intraoperative periprosthetic femoral fractures consisted of 15 nondisplaced calcar cracks (incomplete fractures), 8 greater trochanteric fractures, 4 cortical perforations, and 2 longitudinal diaphyseal nondisplaced cracks. There was 1 intraoperative acetabular fracture (medial wall). All but 1 periprosthetic femoral fracture occurred in patients with a cementless femoral stem, and >80% of postoperative fractures occurred in patients with a non-collared, flat tapered wedge design.

Similar to reoperation, female patients (HR, 1.5 [95% CI, 1.1 to 2.1]; p = 0.04) were significantly more likely to sustain a nonoperative complication relative to male patients, and patients with a BMI of ≥40 kg/m^2 (HR, 2.3 [95% CI, 1.3 to 4.1]; p = 0.018) were significantly more likely to sustain a nonoperative complication relative to patients with a BMI of <40 kg/m^2 (Table III). BMI did not influence the incidence of recurrent instability (HR, 1.05; p = 0.976).

Clinical Outcomes

The mean HHS improved from 53 preoperatively to 94 at the 2-year follow-up and 92 at the 5-year follow-up.

Discussion

DAA THA is gaining in popularity[1]. Although the early experience with this approach was impacted by many surgeons contending with a steep learning curve during adoption[3-5], results after this period[8,14], and among surgeons with training in the approach[23,24], have been encouraging. This approach has been facilitated in contemporary practice by the development of approach-specific instrumentation and implants more amenable to an anterior approach and easier femoral preparation. Given this, modern outcomes after DAA THA are worth exploring. Furthermore, defining contemporary indications leading to revision surgery is of importance.

We found that periprosthetic fracture and superficial wound complication were the most common complications following DAA THA. In our series, postoperative periprosthetic fracture occurred in 1.5% of patients, but only 26 patients (0.7%) required revision or reoperation for this indication. This rate is similar to the rate of postoperative fracture previously reported by Abdel et al. (1.6% at 10 years) in a large cohort of THA cases performed

TABLE III Results of Cox Regression Analyses Comparing Outcomes Based on Patient and Surgeon Characteristics

	Any Revision		Any Reoperation		Nonoperative Complication	
Variable	HR*	P Value†	HR*	P Value†	HR*	P Value†
Sex		0.465‡		**0.012**‡		**0.043**‡
Male	1.0 (reference)		1.0 (reference)		1.0 (reference)	
Female	1.30 (0.79 to 2.13)		1.80 (1.24 to 2.61)		1.50 (1.09 to 2.06)	
Age						
Per additional 10 years	1.15 (0.87 to 1.51)	0.465‡	0.91 (0.75 to 1.10)	0.465‡	0.94 (0.80 to 1.11)	0.542‡
<65 years	1.0 (reference)	0.378§	1.0 (reference)	0.542§	1.0 (reference)	0.869§
≥65 years	1.43 (0.82 to 2.52)		0.86 (0.58 to 1.29)		1.03 (0.73 to 1.45)	
BMI						
Per additional 5 kg/m^2	1.20 (0.99 to 1.46)	0.151‡	1.34 (1.18 to 1.52)	**<0.001**‡	1.13 (0.99 to 1.28)	0.141‡
<40 kg/m^2	1.0 (reference)	0.060#	1.0 (reference)	**<0.001**#	1.0 (reference)	**0.018**#
≥40 kg/m^2	2.73 (1.17 to 6.37)		3.52 (2.01 to 6.18)		2.31 (1.31 to 4.10)	
Surgeon volume		0.542‡		0.230‡		0.869‡
High**	1.0 (reference)		1.0 (reference)		1.0 (reference)	
Low	1.54 (0.48 to 4.96)		1.86 (0.86 to 4.03)		1.08 (0.50 to 2.30)	

*The values are given as the HR, with the 95% CI in parentheses. †Significant values are shown in bold. All p values have been adjusted to control the false discovery rate using the Benjamini-Hochberg method. ‡Based on the multivariable Cox proportional hazards regression model containing sex, age, BMI, Charlson Comorbidity Index (age and severity-weighted), and surgeon volume. §Based on the multivariable Cox proportional hazards regression model containing sex, age (≥65 compared with <65 years), BMI, Charlson Comorbidity Index (age and severity-weighted), and surgeon volume. #Based on multivariable Cox proportional hazards regression model containing sex, age, BMI (≥40 compared with <40 kg/m^2), Charlson Comorbidity Index (age and severity-weighted), and surgeon volume. **Surgeon volume was considered high if the surgeon performed at least 75 cases over the study period.

primarily via posterior or anterolateral approaches[25]. Similar to reports from other authors, we found that collared stem designs are protective against reoperation for periprosthetic fracture[26-28]. Intraoperative fracture occurred in 0.8% of patients in our series. Although 50% of these were nondisplaced calcar cracks managed with cerclage wire, there were 4 cortical perforations (0.1%). Our reported intraoperative fracture rate is similar to that reported by Sershon et al.[29] (0.6%) but lower than the rate previously reported by Abdel et al.[25] (1.7%) in a large series of primary THAs. Although we did have 4 cases (0.1%) of intraoperative femoral cortical perforation, this is lower than the 0.4% rate reported by Kinney et al.[30].

Our series had 98 cases (2.7%) of superficial wound complications, of which 42 required irrigation and debridement. Although this complication has been reported to occur in up to 12% of cases by some authors, most series have reported wound complication rates between 1% and 3%[31-36]. It is reassuring that, despite this rate of superficial wound complications, infectious sequelae were uncommon[37] and our 5-year cumulative incidence of PJI (0.9%) is in line with other contemporary series and registry data[19,31]. We previously reported that female sex and higher BMI were associated with the development of superficial dehiscence following DAA THA[37]. Given that superficial wound complication was a leading indication for reoperation as well as the most common postoperative complication treated nonoperatively, it is perhaps not surprising that we found that female sex and BMI of ≥40 kg/m^2 were risk factors for these 2 outcomes.

Recent literature has demonstrated that, in the United States, hip instability (28%), aseptic loosening (24%), PJI (18%), and periprosthetic fracture (18%) continue to be the leading indications for revision THA when including all surgical approaches[38]. This is in contrast to the findings in the current investigation, in which instability accounted for only 6% of revision cases. Following primary DAA THA in our series, recurrent instability occurred in 12 patients (0.3%), of whom 4 (0.1%) required revision. Similarly, Haynes et al.[14] reported that 0.5% of their 5,065 patients undergoing DAA THA sustained a dislocation and 0.2% required revision for instability. They found that these rates were significantly lower than their institutional dislocation rate using a posterolateral approach[14]. Similar results have been previously reported from institutional[18] and international registries[17,19]. Although the current investigation did not compare approaches, our inclusion of only DAA THA likely accounts for our discordance with the national U.S. data[38].

We found that the most frequent indications for revision were PJI, periprosthetic fracture, and aseptic loosening. Together, these 3 indications accounted for 90% of revision procedures. PJI and periprosthetic fracture, as mentioned, are known to be common indications for revision surgery[38]. Aseptic loosening, particularly femoral aseptic loosening, has previously been noted to be more common following DAA THA[17,19]. In our series, femoral aseptic loosening occurred in nearly 4 times more patients (11) than acetabular aseptic loosening (3). Although aseptic loosening was a leading indication for revision, revision for aseptic loosening

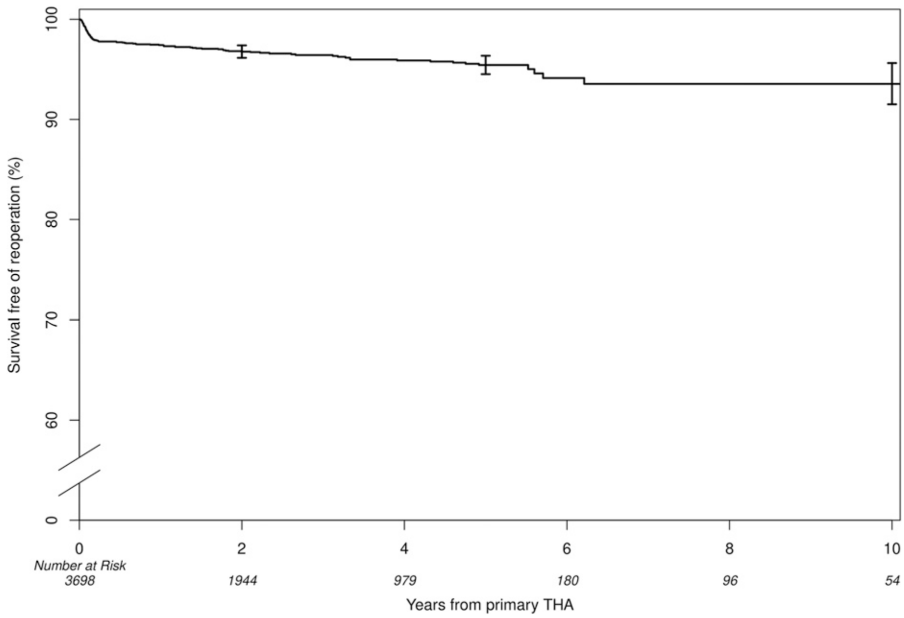

Fig. 2
Estimated survivorship free of all reoperations following primary DAA THA was 97% at 2 years, 95% at 5 years, and 94% at 10 years postoperatively. The error bars indicate the 95% CIs.

was still uncommon and the 5-year cumulative incidence of aseptic loosening leading to revision was 0.84%.

We found that a BMI of ≥40 kg/m^2 was a risk factor for PJI after DAA THA. Despite accounting for <4% of the included patient population, patients with a BMI of ≥40 kg/m^2 accounted for 21% of PJI cases observed in this series. This finding is in concordance with prior data from our institution demonstrating higher rates of PJI with increasing BMI[39]. Similarly, in an analysis of patients undergoing DAA THA, Purcell et al. found that patients with a BMI of ≥35 kg/m^2 had increased rates of PJI[40]. Those authors reported a PJI rate of 2.5% in obese patients[40], which is similar to our rate of 3.6% in a similar population given our longer follow-up and analysis of patients with a BMI of ≥40 kg/m^2.

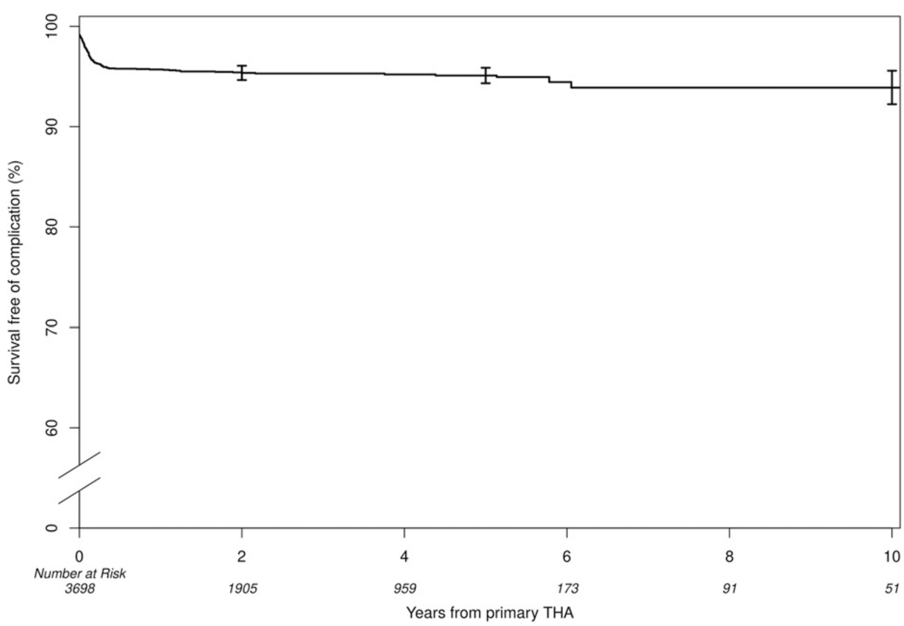

Fig. 3
Kaplan-Meier survivorship curve demonstrating survivorship free of nonoperative complications after primary DAA THA. Complications primarily occurred early, driven by intraoperative periprosthetic fracture and superficial wound complications. The error bars indicate the 95% CIs.

There were multiple limitations to the present study. First, it was a retrospective review of cases from a single health-care system. Although we included patients from 3 geographically distinct institutions and 23 contributing surgeons, it should be recognized that 95% of included cases were performed by 1 of 8 surgeons. Furthermore, despite our analysis not demonstrating a difference in outcomes based on surgeon volume, it should be acknowledged that the surgeons included in this series are arthroplasty specialists, and these results may not be generalizable to surgeons without specialized training in hip arthroplasty. It should also be noted that our analysis of surgeon volume was likely underpowered as a result of overall low event rates. Second, we could not eliminate some selection bias in patient selection for DAA THA. For instance, some surgeons changed the surgical approach, favoring a laterally based approach, in patients with high BMI or for more technically complex cases. Third, although we presented a contemporary series with a relatively large number of revision cases following DAA THA, some indications for revision, such as trunnion corrosion, should be less common with current implants and nearly universal use of ceramic femoral heads[41]. Fourth, we did not perform a radiographic analysis on all included patients. It is possible that some patients may have had radiolucencies indicative of future failure, but they were not captured in our presented numbers. Lastly, we presented the mean 4-year follow-up data, and monitoring outcomes over longer follow-up is warranted.

In conclusion, DAA THA is associated with high revision and reoperation-free survivorship at early to intermediate follow-up. We found that leading indications for revision following DAA THA were PJI, periprosthetic fracture, and aseptic loosening. Importantly, our data demonstrated that failure due to hip instability is uncommon and this complication occurs in a small minority of patients after DAA THA. As a note of caution, we report that patients with a BMI of ≥40 kg/m^2 had higher rates of reoperation and complications, including PJI. Future work should continue to monitor these patients over longer-term follow-up. ■

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